Actualized Geometry and a Null Prediction for Gravity-Mediated Entanglement

The Bose–Marletto–Vedral Protocol in Relational Actualism

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Abstract

We show that Relational Actualism (RA) predicts a strict null result for gravity-mediated entanglement in the Bose–Marletto–Vedral (BMV) protocol. The prediction follows from a single architectural principle: the effective spacetime metric is sourced only by the actualized causal ledger—the growing directed acyclic graph of irreversible physical events—not by unactualized quantum branch structure. During coherent spatial superposition, the positional alternatives of the test masses remain in the reversible possibility layer (Step 2 of the Engine of Becoming) and have not yet been inscribed as actualized vertices (Step 4). Since the metric is recalculated only from actualized vertices (Step 5), no branch-resolved geometry exists during the coherent evolution interval. The gravitational phase is therefore common-mode across all positional branches, and the entangling invariant vanishes identically: $\Delta\phi_{\text{grav}} = 0$. We prove a coherence–geometry incompatibility theorem showing that the two conditions required by the standard BMV mechanism—maintained positional coherence and branch-resolved gravitational geometry—are structurally mutually exclusive in RA. A positive BMV signal, after exclusion of nongravitational channels, would directly falsify the Step-2/Step-5 separation that is central to the RA architecture.

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1 Introduction

The Bose–Marletto–Vedral (BMV) protocol (1; 2) proposes to test whether gravity can mediate quantum entanglement between two mesoscopic masses, each placed in a coherent spatial superposition. If gravity is fundamentally quantum, the branch-dependent Newtonian interaction generates a nonseparable phase that entangles the masses. If gravity is fundamentally classical or—as Relational Actualism (RA) holds—fundamentally *actualized*, no such entanglement arises.

The standard BMV argument implicitly assumes that each branch of the spatial superposition sources its own gravitational field prior to any measurement or actualization event. In the language of RA, this amounts to assuming that the effective metric responds to the Level-2 quantum-measure amplitude structure—the space of unrealized possibilities—rather than exclusively to the Level-3 actualized causal ledger.

RA denies precisely this assumption. The framework's central architectural feature is a sharp separation between three levels of physical description:

- **Step 2 (Propose):** Reversible quantum amplitudes are assigned to candidate continuations of the actualized graph. No vertices are written.
- **Step 4 (Inscribe):** One candidate is selected and irreversibly written into the causal DAG as a new actualized vertex, subject to the BDG filter $S_{\mathrm{BDG}} > 0$.
- **Step 5 (Update geometry):** The effective metric $g_{\mu\nu}$ is recalculated from the updated actualized graph via the coarse-graining map $g_{\mu\nu} = \mathcal{G}[G_n]$.

The BMV protocol requires Step 5 to respond at Step 2. RA says Step 5 responds only at Step 4. That is the entire content of the null prediction, and this note makes it precise.

This paper does and does not:

✓ Proves a null BMV theorem from RA's Step-2/Step-5 separation.
✓ States the precise falsification condition for RA.
✓ Distinguishes RA's prediction from both quantized gravity and semiclassical gravity.
× Does not derive the full RA ontology (see companion papers (3; 4; 5)).
× Does not claim gravity is "classical" in the naive sense.

2 The Standard BMV Mechanism

Two equal masses A and B, each of mass m, are prepared in the spatial superposition

$$|+\rangle_X = \frac{1}{\sqrt{2}}\big(|L\rangle_X + |R\rangle_X\big), \qquad X \in \{A, B\}. \tag{1}$$

The initial joint state is

$$|\Psi_0\rangle = |+\rangle_A \otimes |+\rangle_B = \frac{1}{2}\big(|LL\rangle + |LR\rangle + |RL\rangle + |RR\rangle\big). \tag{2}$$

Each branch $|ij\rangle$ has separation d_{ij} and Newtonian interaction energy $U_{ij} = -Gm^2/d_{ij}$. After interaction time t, branch $|ij\rangle$ accumulates the gravitational phase

$$\phi_{ij} = \frac{Gm^2 t}{\hbar\, d_{ij}}. \tag{3}$$

The evolved state is

$$|\Psi_{\text{std}}(t)\rangle = \frac{1}{2} \sum_{i,j\in\{L,R\}} e^{i\phi_{ij}} |ij\rangle. \tag{4}$$

This state is entangled whenever the entangling phase invariant

$$\Delta\phi := \phi_{LL} + \phi_{RR} - \phi_{LR} - \phi_{RL} \tag{5}$$

is nonzero modulo 2π. For generic separations, $\Delta\phi \neq 0$, and the standard prediction is positive gravity-mediated entanglement.

The implicit assumption behind this calculation is that each branch $|ij\rangle$ has its own gravitational potential, sourced by the mass configuration that branch represents. That is the assumption RA removes.

3 RA Architecture: The Step-2/Step-5 Separation

In Relational Actualism, the universe is represented by a growing causal directed acyclic graph (DAG) $G = (V, \prec)$ whose vertices are actualized events and whose edges encode causal precedence. The dynamics proceeds through the five-step Engine of Becoming:

1. **Survey:** Identify the boundary of G (vertices with no future-directed edges).
2. **Propose:** Assign complex amplitudes to candidate extensions of G, weighted by the BDG action $S = 1 - N_1 + 9N_2 - 16N_3 + 8N_4$.
3. **Filter:** Retain only candidates with $S_{\text{BDG}} > 0$.
4. **Inscribe:** Select one candidate and write it irreversibly into the graph as a new vertex.
5. **Update:** Recalculate the effective metric from the updated actualized graph: $g_{\mu\nu} = \mathcal{G}[G]$.

The critical architectural feature for BMV is the separation between Steps 2 and 5. Step 2 creates a weighted space of possibilities. Step 5 responds only to what has been inscribed at Step 4.

Definition 3.1 (Ledger-sourced metric)**.** *Let $G_n = (V_n, E_n)$ be the actualized causal DAG after n actualization events. The emergent metric at stage n is*

$$g_{\mu\nu}^{(n)} = \mathcal{G}[G_n], \tag{6}$$

where $\mathcal{G}$ is the Step-5 coarse-graining map. This metric depends only on actualized vertices and edges, not on unactualized candidate extensions present in the Step-2 amplitude layer.

Definition 3.2 (Branch-coherent regime)**.** *A BMV protocol is in the* branch-coherent regime *on the interval $[0, T]$ if, for each mass, the positional alternatives L and R remain in coherent superposition throughout $[0, T]$: no irreversible positional actualization event occurs during that interval. Equivalently, the positional degree of freedom remains at Step 2 rather than passing through Step 4.*

4 The RA-BMV Null Theorem

Lemma 4.1 (Step-2 non-sourcing)**.** *Let $\mathcal{C}_n$ denote the set of candidate extensions available at Step 2 from ledger state G_n. Suppose a degree of freedom remains fully within the Step-2 amplitude layer during $[0, T]$, so that none of its alternatives are selected at Step 4 during that interval. Then the Step-5 metric is insensitive to the decomposition of amplitudes across those alternatives.*

In symbols: if no candidate in the relevant sector actualizes, then for that sector

$$G(t) = G_n \quad \textit{for all } t \in [0, T], \tag{7}$$

and therefore

$$g_{\mu\nu}(t) = \mathcal{G}[G_n] \quad \textit{for all } t \in [0, T]. \tag{8}$$

Proof. Step 2 enumerates weighted possibilities; it does not add vertices to the actualized graph. By hypothesis, no positional branch is selected at Step 4 during $[0, T]$. Therefore the actualized graph in the relevant sector does not change. Since Step 5 depends only on the actualized graph (Definition 3.1), the metric remains unchanged throughout the interval. □

Lemma 4.2 (No branch-indexed metric before actualization)**.** *In the branch-coherent regime, there is no ontically meaningful family*

$$\{g_{\mu\nu}^{LL},\ g_{\mu\nu}^{LR},\ g_{\mu\nu}^{RL},\ g_{\mu\nu}^{RR}\} \tag{9}$$

of simultaneously available branch-resolved metrics.

Proof. Such a family would require the four branch labels LL, LR, RL, RR to correspond to four distinct actual source configurations. But in the branch-coherent regime, those labels denote only unactualized Step-2 alternatives. By Lemma 4.1, Step 5 has access only to the common actual ledger, not to branch labels that have not passed through Step 4. □

Theorem 4.1 (RA-BMV null theorem)**.** *Consider a BMV protocol with two masses A and B, each prepared in a coherent two-position superposition. Assume:*

1. *the protocol remains in the branch-coherent regime during the interaction interval $[0, T]$;*
2. *the emergent metric is Step-5 sourced from the actualized ledger only;*
3. *nongravitational entangling channels are absent.*

Then the purely gravitational contribution to the four branch phases is common-mode:

$$\phi_{LL}^{\text{grav}} = \phi_{LR}^{\text{grav}} = \phi_{RL}^{\text{grav}} = \phi_{RR}^{\text{grav}}, \tag{10}$$

and the entangling phase invariant vanishes:

$$\boxed{\Delta\phi_{\text{grav}} := \phi_{LL} + \phi_{RR} - \phi_{LR} - \phi_{RL} = 0.} \tag{11}$$

Hence the protocol cannot generate gravity-mediated entanglement.

Proof. By Lemma 4.2, no branch-resolved metric exists during the coherent interval. Gravity contributes the same Step-5 background potential $U_0 = U[g_{\mu\nu}(G_n)]$ to each amplitude component. The induced gravitational phase is identical on all four branches: $\phi_{ij}^{\text{grav}} = -U_0 t/\hbar$ for all i, j. Therefore

$$|\Psi_{\text{RA}}(t)\rangle = e^{-iU_0 t/\hbar}\frac{1}{2}\big(|LL\rangle + |LR\rangle + |RL\rangle + |RR\rangle\big) = e^{-iU_0 t/\hbar}\,|+\rangle_A \otimes |+\rangle_B. \tag{12}$$

The state remains separable, so $\Delta\phi_{\text{grav}} = 0$. □

5 Coherence–Geometry Incompatibility

The null result is not contingent on experimental parameters such as mass, separation, or interaction time. It reflects a structural incompatibility between the two conditions the standard BMV mechanism requires.

Theorem 5.1 (Coherence–geometry incompatibility)**.** *In RA, the following two conditions cannot simultaneously hold for the same positional degree of freedom over the same interval:*

1. Branch coherence: *the alternatives L and R remain in coherent superposition.*
2. Branch-resolved geometry: *the alternatives L and R source distinct effective metrics.*

Proof. If branch coherence holds, the alternatives remain at Step 2 and have not entered the ledger; by Lemma 4.1 they cannot source distinct Step-5 metrics. Conversely, if branch-resolved geometry exists, the branch label must already be represented in the actualized ledger, meaning Step 4 has occurred and coherence is lost. □

This is the deeper structural reason for the null prediction. The standard BMV mechanism requires simultaneous access to both the coherent superposition *and* the branch-dependent geometry that would produce the entangling phase. In RA, these are architecturally mutually exclusive.

6 What "Spacetime Superposition" Means in RA

The standard BMV framework implicitly represents the evolved state as a superposition of matter–geometry composites:

$$\frac{1}{2}\big(|LL\rangle|g_{LL}\rangle + |LR\rangle|g_{LR}\rangle + |RL\rangle|g_{RL}\rangle + |RR\rangle|g_{RR}\rangle\big). \tag{13}$$

RA replaces this with

$$\left(\frac{1}{2}\sum_{i,j\in\{L,R\}}|ij\rangle\right)\otimes|g[\mathcal{G}_n]\rangle. \tag{14}$$

The phrase "spacetime superposition" must therefore be disambiguated:

- **Forbidden in RA:** A superposition of presently actual effective metrics—Eq. (13).
- **Permitted in RA:** A superposition of Step-2 growth channels whose later actualization would yield different Step-5 metrics—i.e., superposition over geometry-producing *futures*, not over the presently actual geometry.

This distinction is not interpretive decoration. It is the architectural content of the Step-2/Step-5 separation and the source of the null prediction.

7 Relation to Competing Frameworks

It is important to distinguish the RA null prediction from two superficially similar positions.

7.1 Quantized gravity

In frameworks where gravity is a quantized field, the metric is a quantum degree of freedom that can be placed in superposition. Each branch of the matter superposition sources its own branch of the metric superposition. The standard BMV mechanism operates, and the prediction is positive: $\Delta\phi \neq 0$.

RA disagrees. The metric is not a quantum degree of freedom. It is the macroscopic record of actualized history.

7.2 Semiclassical gravity

In semiclassical gravity, the metric is sourced by the expectation value of the stress-energy tensor: $G_{\mu\nu} = 8\pi G\,\langle\hat{T}_{\mu\nu}\rangle$. A spatial superposition would produce an averaged field that is neither the $|L\rangle$ nor the $|R\rangle$ configuration but a weighted mean. The BMV prediction in this framework is ambiguous: some treatments predict partial entanglement, others predict none.

RA disagrees with semiclassical gravity as well. The source is not the expectation value of unactualized branch structure. The source is the actualized ledger alone:

$$G_{\mu\nu} = 8\pi G\,P_{\text{act}}[T_{\mu\nu}], \tag{15}$$

where P_{act} is the actualization projector that retains only on-shell, ledger-written contributions.

Thus RA's null result is not "gravity is classical." It is specifically *gravity is actualized*: sourced only by irreversible causal facts, not by reversible quantum possibilities.

8 Experimental Consequences

8.1 The prediction

In an ideal BMV experiment with all nongravitational channels excluded, RA predicts:

$$\Delta\phi^{\text{RA}}_{\text{grav}} = 0, \tag{16}$$

and any entanglement monotone attributed solely to gravity must vanish:

$$\mathcal{E}_{\text{grav}}^{\text{RA}} = 0. \tag{17}$$

8.2 What RA allows

RA does not predict the complete absence of all experimental signatures. The following effects may be present:

- Common-mode gravitational phases (global, nonfactorizable contribution zero).
- Environmental decoherence from nongravitational channels.
- Electromagnetic or Casimir contamination.
- Trap-dependent systematic phases.

The prediction is that after removal of these nongravitational contributions, no purely gravitational entangling signal remains.

8.3 Falsification condition

A positive BMV result—specifically, a nonzero entanglement witness attributed to gravity after exclusion of all nongravitational channels—would directly falsify the Step-2/Step-5 separation of the Engine of Becoming. It would demonstrate that the effective metric responds to unactualized quantum branch structure, collapsing the distinction between possibility and actuality that is central to RA.

The logic is unusually clean:

- **Null BMV:** consistent with RA.
- **Positive BMV:** incompatible with present-form RA.

This makes the BMV experiment one of the sharpest near-term empirical discriminators for the RA programme.

9 Discussion

The RA-BMV null prediction reveals something deep about the framework. RA does not merely classify gravity as classical or quantum. It makes a more specific structural claim: *gravity is the macroscopic description of what has already become actual.*

The metric $g_{\mu\nu} = \mathcal{G}[G_n]$ is not a fundamental field. It is the large-scale order of the actualized causal ledger. Asking whether gravity can be "in superposition" is, in RA, a category error: you cannot superpose the record of what has happened, because what has happened is by definition actual and definite.

This does not mean RA denies quantum phenomena in gravitational settings. Gravitational effects such as gravitational redshift, time dilation, and frame-dragging are all present in RA as consequences of the actualized graph's large-scale structure. What RA denies is that the metric itself can be placed in a coherent superposition of branch-resolved configurations, because that would require the actualized ledger to contain mutually exclusive entries simultaneously.

The BMV experiment is therefore not merely a test of whether gravity is "quantum enough" to mediate entanglement. It is a test of whether the effective spacetime geometry belongs to the space of actualized facts or to the space of unresolved possibilities. RA predicts the former. A positive BMV result would require the latter.

10 Conclusion

We have shown that Relational Actualism predicts a strict null result for gravity-mediated entanglement in the BMV protocol. The prediction follows from a single principle: the effective metric is sourced only by the actualized causal ledger, not by unactualized quantum branch structure. The coherence–geometry incompatibility theorem (Theorem 5.1) establishes that the two conditions required by the standard BMV mechanism—maintained positional coherence and branch-resolved gravitational geometry—are structurally mutually exclusive in RA.

Multiple experimental groups are currently developing BMV-type experiments (1; 2). A null result would be consistent with RA and would constrain frameworks in which gravity is a quantized degree of freedom. A positive result would directly falsify the Step-2/Step-5 separation that is the architectural foundation of Relational Actualism.

Either outcome advances physics.

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